

Analyzing data on the level of cognitive processes - ESCoP 2025

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Exercise 1: Central Limit Theorem

The central limit theorem states that the sum or average of random samples from any probability distribution with a mean and standard deviation smaller than infinity will approximate a normal distribution, with a sufficient number of samples. This exercise should help you getting an intuition what a “sufficient number” of samples is and what this implies for the assumption of the normal distribution in experimental settings.

Your task is to:

1. Choose any - ideally asymmetric or totally random (e.g. using the `sample` function) - probability distribution.
2. Compute the average of 2, 5, 10, and 20 samples from the distribution for 200 times.
3. Plot these averages in a histogram and try overlaying the density of the normal distribution with the mean and standard deviation of the plotted values